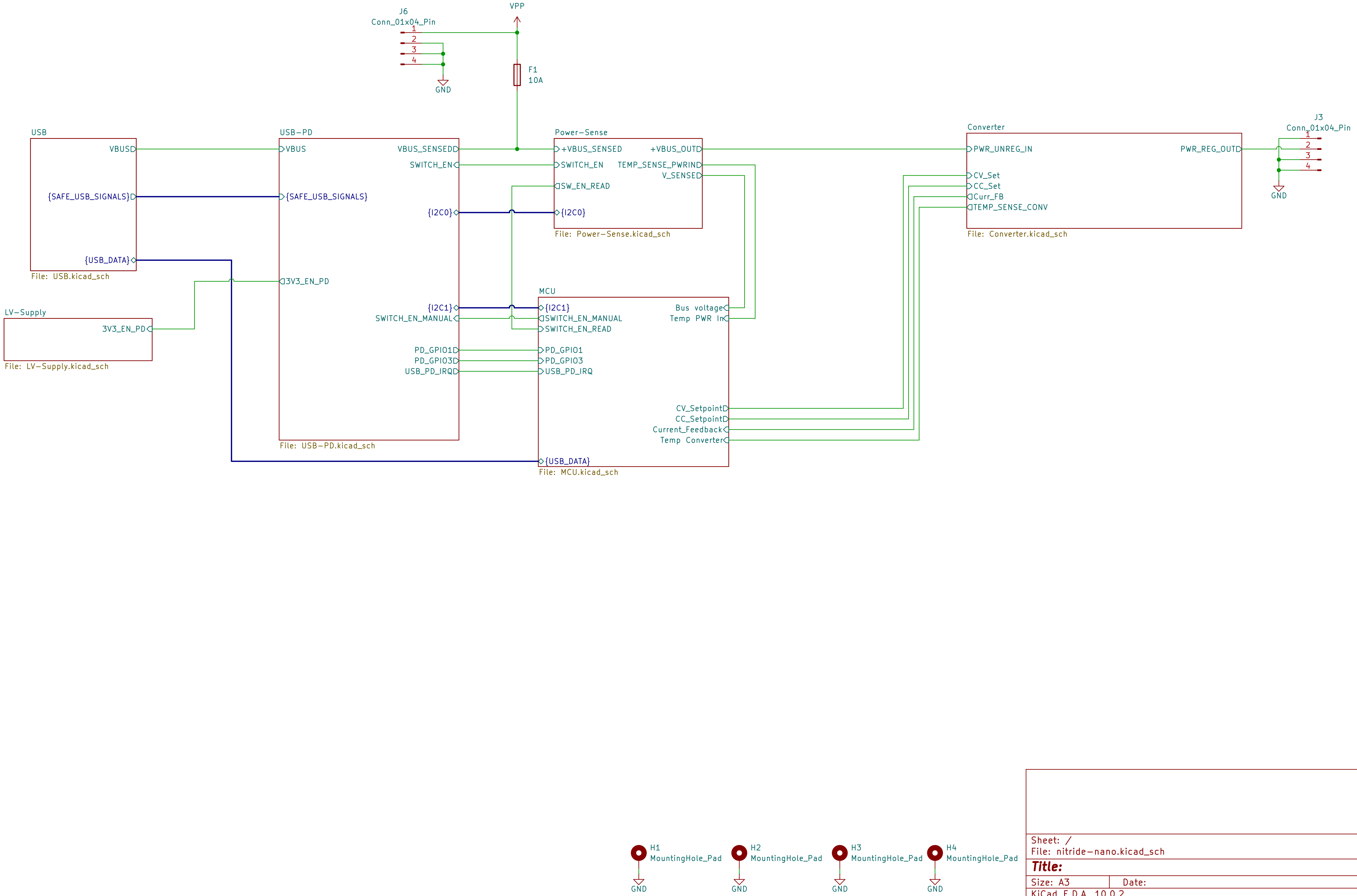

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Size: A3 | Date: | **Rev:**

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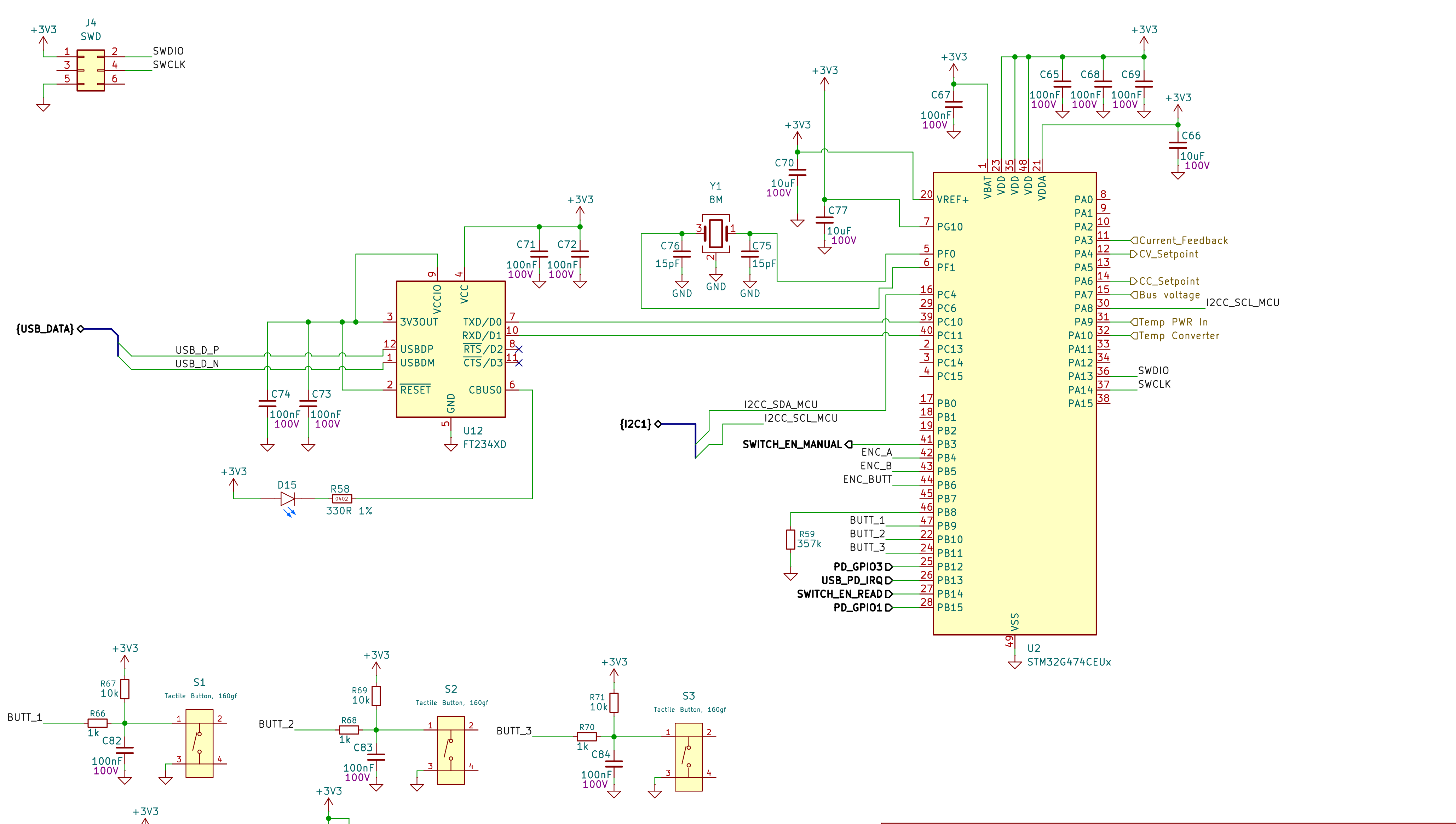

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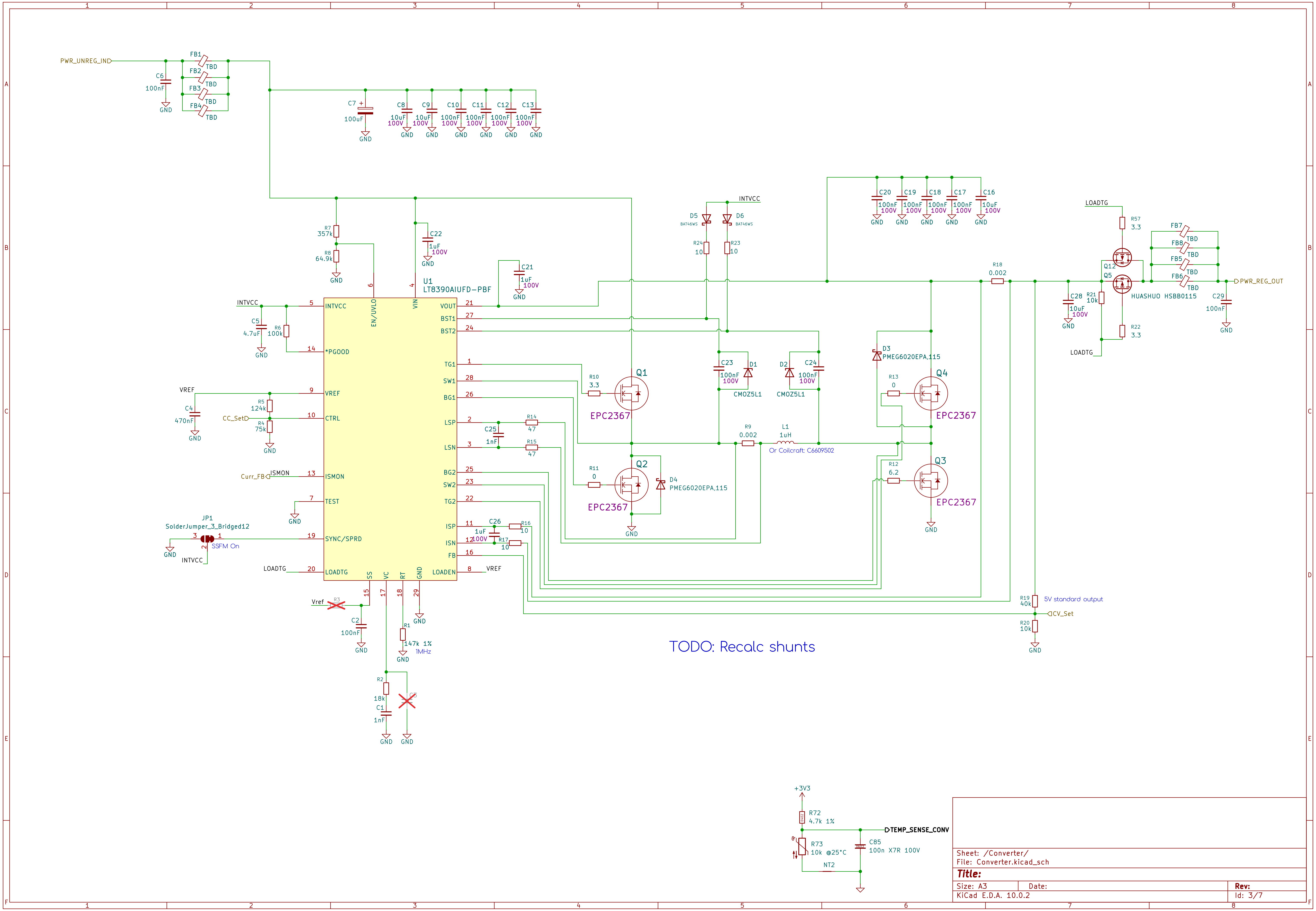

PWR_UNREG_IN
C6 100nF
FB1 TBD
FB2 TBD
FB3 TBD
FB4 TBD
C7 100uF
C8 10uF 100V
C9 10uF 100V
C10 100nF 100V
C11 100nF 100V
C12 100nF 100V
C13 100nF 100V
GND
R7 357k
R8 64.9k
C22 1uF 100V
C21 1uF 100V
U1 LT8390AIUFD-PBF
INTVCC
C5 4.7uF
R6 100k
INTVCC
*PGOOD
EN/UVLO
VIN
VOUT
BST1
BST2
TG1
SW1
BG1
LSP
LSN
BG2
SW2
TG2
ISP
ISN
FB
LOADEN
VREF
C4 470nF
R5 124k
CC_Set
CTRL
R4 75k
Curr_FB
ISMON
TEST
JP1 SolderJumper_3_Bridged12
SSFM On
SYNC/SPRD
LOADTG
SS
VC
RT
Vref
R3
C2 100nF
R1 147k 1% 1MHz
R2 18k
C1 1nF
C3
C25 1nF
R14 47
R15 47
C26 1uF 100V
R16 10
R17 10
D5 BAT46WS
D6 BAT46WS
R24 10
R23 10
C23 100nF 100V
D1 CMOZ5L1
D2 CMOZ5L1
C24 100nF 100V
R10 3.3
Q1 EPC2367
R11 0
Q2 EPC2367
D4 PMEG6020EPA,115
R9 0.002
L1 1uH
Or Coilcraft: C6609502
D3 PMEG6020EPA,115
R13 0
Q4 EPC2367
R12 6.2
Q3 EPC2367
C20 100nF 100V
C19 100nF 100V
C18 100nF 100V
C17 100nF 100V
C16 10uF 100V
R18 0.002
LOADTG
R57 3.3
FB7 TBD
FB8 TBD
FB5 TBD
FB6 TBD
Q12
Q5
HUASHUO HSBB0115
R21 10k
R22 3.3
C28 10uF 100V
C29 100nF
PWR_REG_OUT
R19 40k
5V standard output
CV_Set
R20 10k
TODO: Recalc shunts
+3V3
R72 4.7k 1%
TEMP_SENSE_CONV
R73 10k @25°C
C85 100n X7R 100V
NT2
Sheet: /Converter/
File: Converter.kicad_sch
Title:
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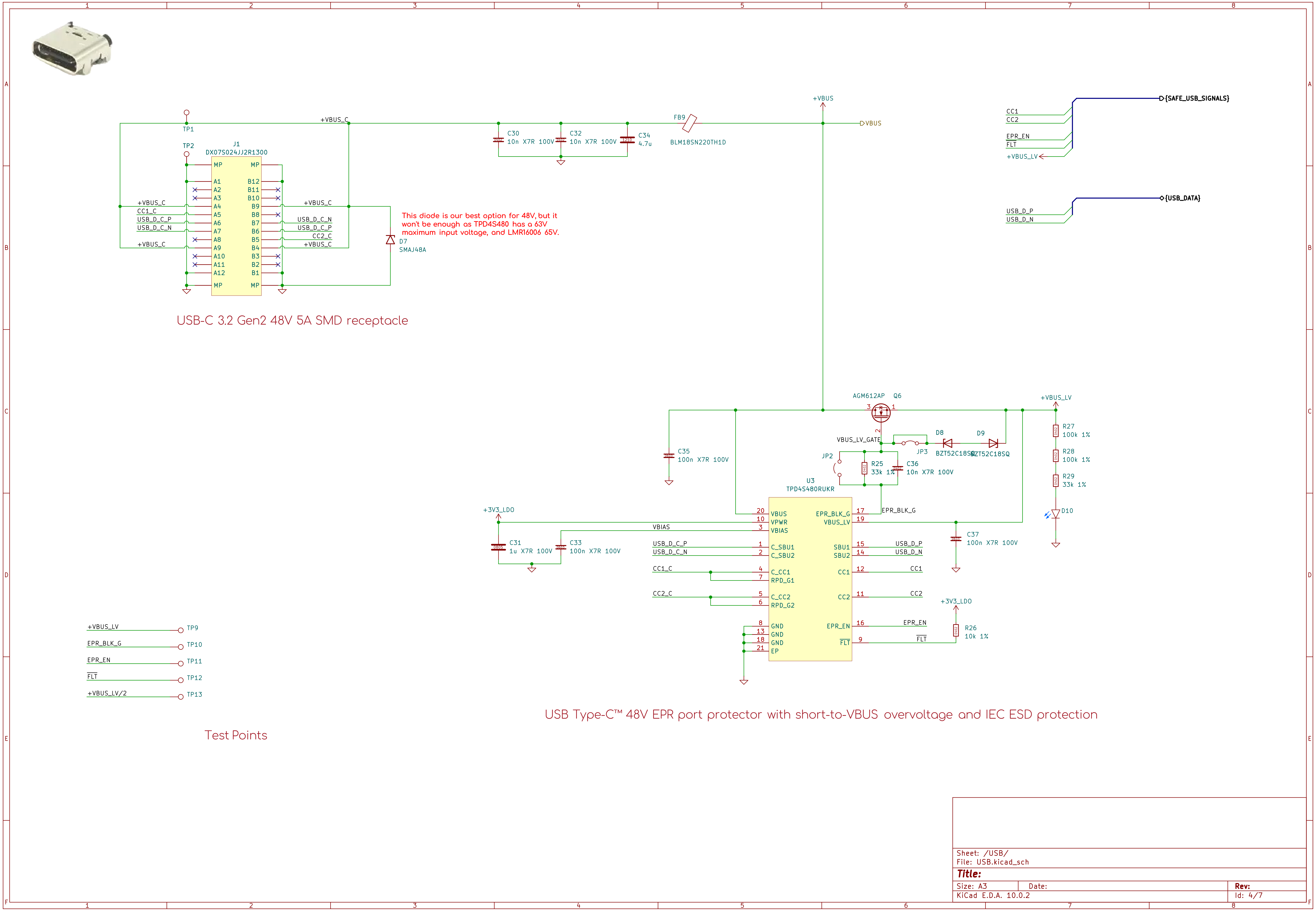

USB-C 3.2 Gen2 48V 5A SMD receptacle
J1
DX07S024JJ2R1300
TP1
TP2
+VBUS_C
CC1_C
USB_D_C_P
USB_D_C_N
CC2_C
This diode is our best option for 48V, but it won't be enough as TPD4S480 has a 63V maximum input voltage, and LMR16006 65V.
D7
SMAJ48A
C30
10n X7R 100V
C32
10n X7R 100V
C34
4.7u
FB9
BLM18SN220TH1D
+VBUS
VBUS
CC1
CC2
EPR_EN
FLT
+VBUS_LV
{SAFE_USB_SIGNALS}
USB_D_P
USB_D_N
{USB_DATA}
AGM612AP
Q6
VBUS_LV_GATE
JP2
JP3
D8
D9
BZT52C18SQ
BZT52C18SQ
R25
33k 1%
C36
10n X7R 100V
C35
100n X7R 100V
R27
100k 1%
R28
100k 1%
R29
33k 1%
D10
U3
TPD4S480RUKR
VBUS
VPWR
VBIAS
C_SBU1
C_SBU2
C_CC1
RPD_G1
C_CC2
RPD_G2
GND
GND
GND
EP
EPR_BLK_G
VBUS_LV
SBU1
SBU2
CC1
CC2
EPR_EN
FLT
+3V3_LDO
C31
1u X7R 100V
C33
100n X7R 100V
C37
100n X7R 100V
R26
10k 1%
USB Type-C™ 48V EPR port protector with short-to-VBUS overvoltage and IEC ESD protection
+VBUS_LV
TP9
EPR_BLK_G
TP10
EPR_EN
TP11
FLT
TP12
+VBUS_LV/2
TP13
Test Points
Sheet: /USB/
File: USB.kicad_sch
Title:
Size: A3
Date:
Rev:
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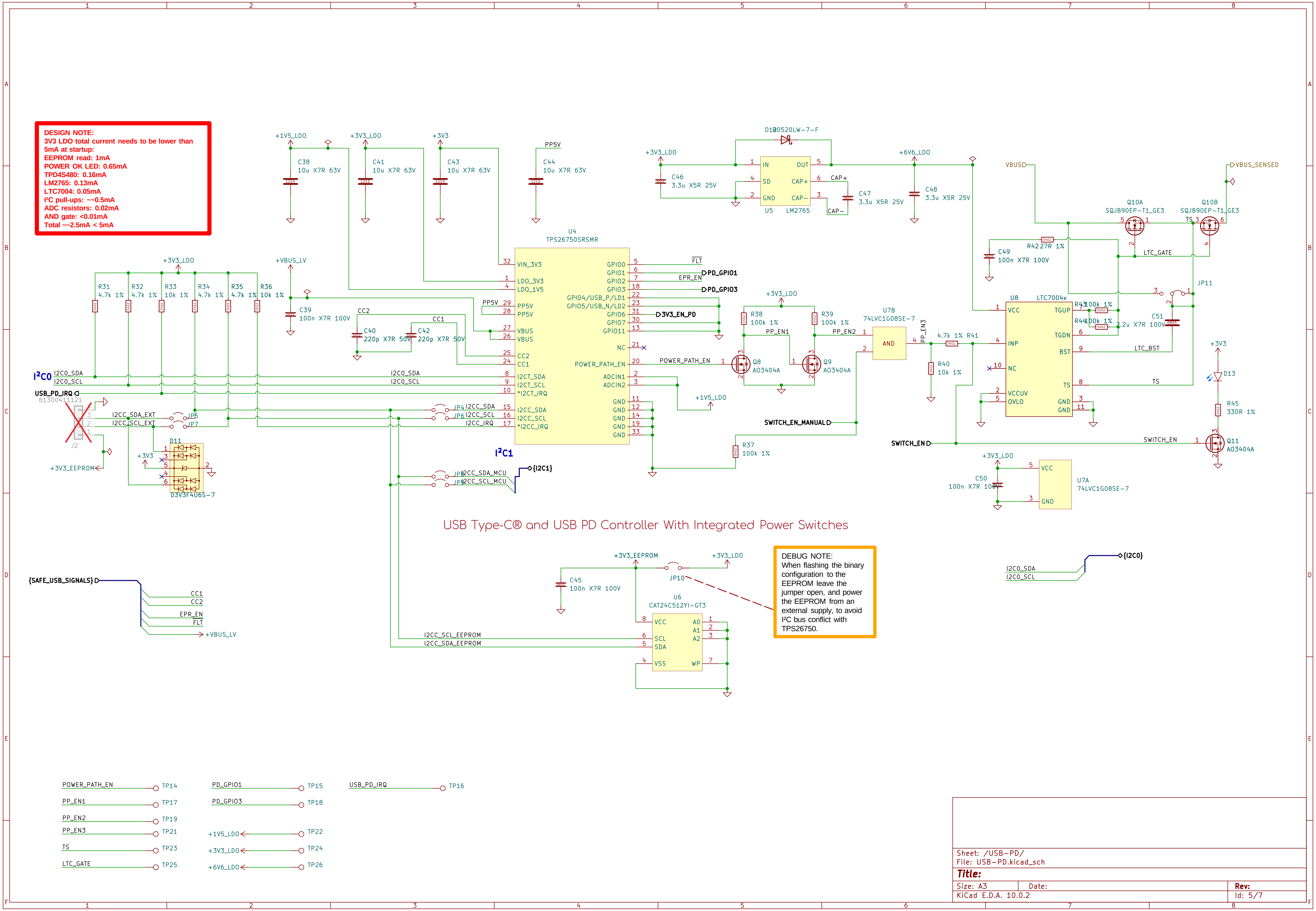

DESIGN NOTE:
3V3 LDO total current needs to be lower than 5mA at startup:
EEPROM read: 1mA
POWER OK LED: 0.65mA
TPD4S480: 0.16mA
LM2765: 0.13mA
LTC7004: 0.05mA
I²C pull-ups: ~~0.5mA
ADC resistors: 0.02mA
AND gate: <0.01mA
Total ~~2.5mA < 5mA
U4
TPS26750SRSMR
U5 LM2765
U8 LTC7004x
U7B 74LVC1G08SE-7
U7A 74LVC1G08SE-7
U6 CAT24C512YI-GT3
Q10A SQJB90EP-T1_GE3
Q10B SQJB90EP-T1_GE3
Q8 AO3404A
Q9 AO3404A
Q11 AO3404A
D11 D3V3F4U6S-7
D13
J2 61300411121
I²C0
I²C1
USB Type-C® and USB PD Controller With Integrated Power Switches
DEBUG NOTE:
When flashing the binary configuration to the EEPROM leave the jumper open, and power the EEPROM from an external supply, to avoid I²C bus conflict with TPS26750.
{SAFE_USB_SIGNALS}
Sheet: /USB-PD/
File: USB-PD.kicad_sch
Title:
Size: A3
Date:
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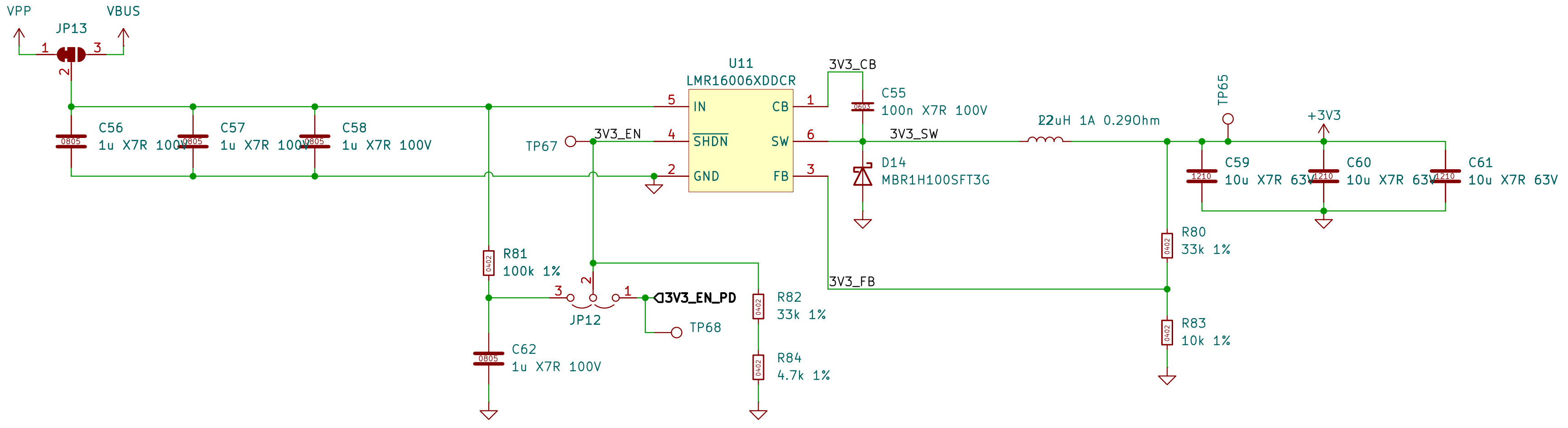

3.3V 600mA 700kHz Buck Regulator

Sheet: /LV-Supply/
File: LV-Supply.kicad_sch

Title:

Size: A4 Date: **Rev:**
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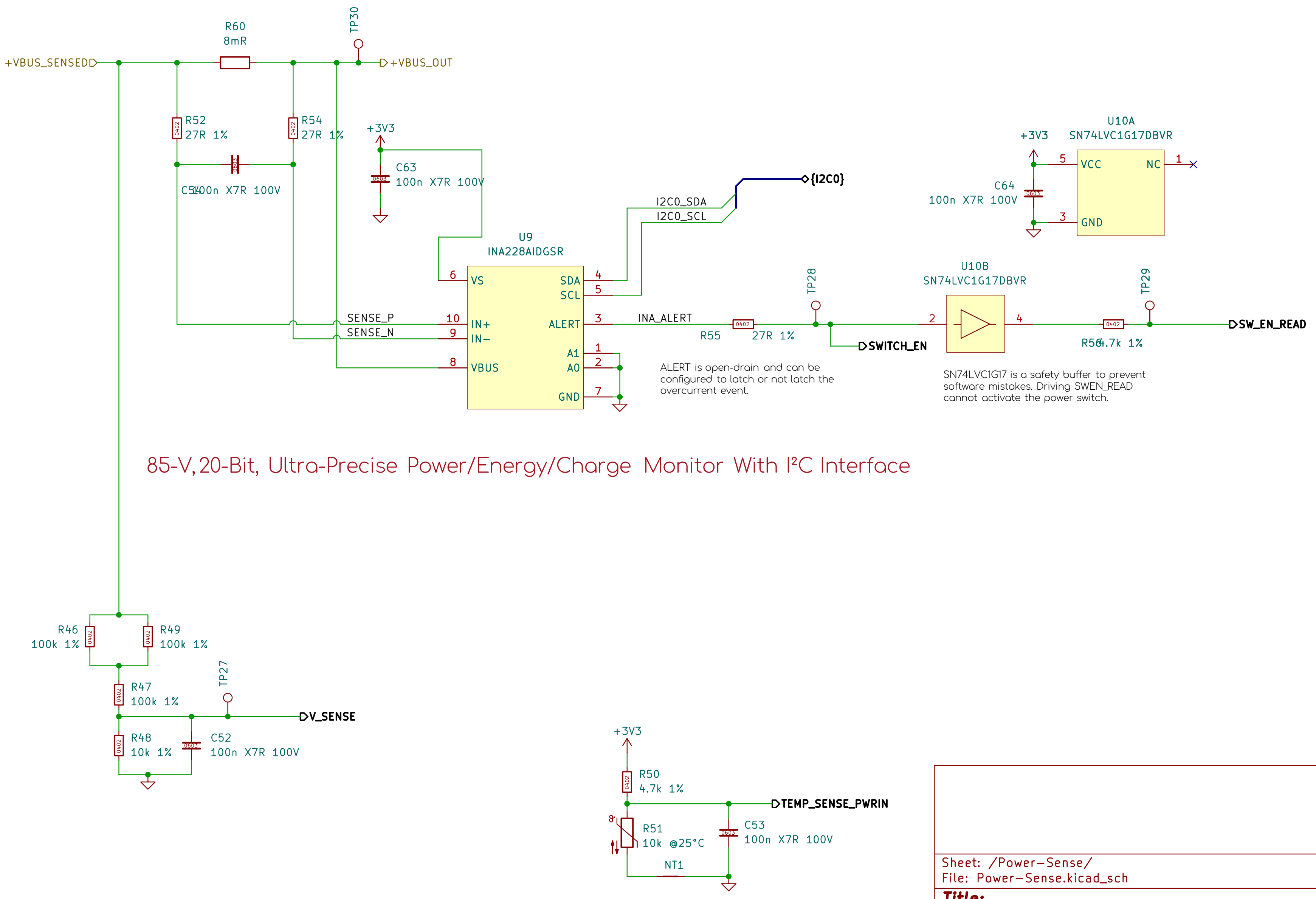

Sheet: /Power-Sense/
File: Power-Sense.kicad_sch

Title:

Size: A4 | Date: | **Rev:**

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